

7(a)	(the ohm is) volt / ampere	B1
7(b)(i)	$R = \rho L / A$	C1
	ratio = $[\rho L / (\pi d^2 / 4)] / [0.028 \rho \times 7.0 L / \{\pi (14d)^2 / 4\}] = 1000$ or ratio = $14^2 / (0.028 \times 7) = 1000$	A1
7(b)(ii)	same current (in connecting and filament wires) and the lamp/filament (wire) has greater resistance	B1
7(b)(iii)	$P = V^2 / R$ or $P = VI$ or $P = I^2 R$	C1
	(for filament wire) $R = 12^2 / 6.0$ or $R = 6.0 / 0.50^2$ or $R = 12 / 0.50$	C1
	(for filament wire) $R = 24 \Omega$	A1
	(for connecting wire) $R = 24 / 1000$ $= 2.4 \times 10^{-2} \Omega$	
7(b)(iv)	resistance of connecting wire increases	B1
	current in circuit/lamp/filament (wire) decreases or potential difference across lamp/filament (wire) decreases	M1
	(so) resistance of lamp/filament (wire) decreases	A1